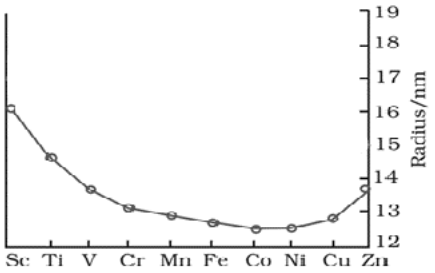
GRADE 12 BOARD PREPARATION REVISION TEST -10

The trend of which property is represented by the following graph?



(a) ionization enthalpy
(b) atomic radii
(c) enthalpy of atomization
(d) melting point

Which of the following is not considered a transition element?

(a) Scandium (b) Silver (c) Vanadium (d) Zinc

. Attempt **any five** of the following:

(a) Which of the following ions will have a magnetic moment value of 1.73 BM.

Sc^{3+}, Ti^{3+}, Ti^{2+}, Cu^{2+}, Zn^{2+}

(b) In order to protect iron from corrosion, which one will you prefer as a sacrificial electrode, Ni or Zn? Why? (Given standard electrode potentials of Ni, Fe and Zn are –0.25 V, –0.44 V and –0.76 V respectively.)

(c) The second ionization enthalpies of chromium and manganese are 1592 and 1509 kJ/mol respectively. Explain the lower value of Mn.

(d) Give two similarities in the properties of Sc and Zn.

(e) What is actinoid contraction? What causes actinoid contraction?

(f) What is the oxidation state of chromium in chromate ion and dichromate ion?

(g) Write the ionic equation for reaction of KI with acidified $KMnO_4$.

Which one of the following element will be strong oxidising agent in their +3 oxidation state? (Atomic no. Cr=24, Mn=25, Ti=22, Fe=26) A. Cr B. Mn C. Ti D. Fe	1
Assertion (A): Permanganate and manganate ions have tetrahedral structure **Reason (R):** In permanganate and manganate ions, pi bonding takes place which is due to overlap of p orbitals of oxygen with p orbitals of manganese. Select the most appropriate answer from the options given below: A. Both A and R are true, and R is the correct explanation of A. B. Both A and R are true, and R is not the correct explanation of A. C. A is true but R is false. D. A is false but R is true.	1
(I) Complete and balance the following reaction $SO_3^{2-} + MnO_4^- + H^+ \rightarrow$	1
(II) Why does titanium melt at a lower temperature than chromium?	1
OR	
(I) Why do titanium and scandium forms complex compounds, whereas potassium and calcium do not, although they belong to same period?	1
(II) Complete and balance the following reaction $Fe^{2+} + Cr_2O_7^{2-} + H^+ \rightarrow$	1
(I) Which of the following has the highest magnetic moment? Justify your answer. Ti^{2+} ,Co^{2+} , Fe^{2+} . [Atomic No. Ti = 22 Co = 27 , Fe = 26]	1
(II) Which characteristic of iron make it suitable to: a) act as catalyst in Haber's process. b) form interstitial compound with oxygen.	2

Answer the following questions: 3x1

I. Are the enthalpies of atomisation of Zinc and Copper matched correctly? Justify your answer.

Element	Enthalpy of atomisation/ $kJmol^{-1}$
Zinc	339
Copper	130

II. Out of sulphuric acid and hydrochloric acid, which acid will you prefer for permanganate titrations and why?

III. $5NO_2^- + 2MnO_4^- + 6H^+ \rightarrow$